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Water resistance and mechanical properties of biopolymer (alginate and soy protein) coated paperboards

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Abstract

The effect of biopolymer coating on the water barrier and mechanical properties of paperboards used as corrugated fiberboard box liners was studied under several conditions. Paperboards were coated with selected biopolymers, such as alginate and soy protein isolate (SPI) and studied with or without further treatment, such as cross-linking through $CaCl_2$ or formaldehyde treatment and compositing with organically modified montmorillonite (OMMT). Biopolymer coating increased the thickness of paperboards from 9% to 16%, depending on the coating materials and treatment methods, and resulted in a smoother and more homogeneous surface. Though the tensile strength (TS) of the coated paperboards decreased from 12.5% to 37.5% of the uncoated paperboards by coating, ring crush strength was not decreased. Wetting properties, such as contact angle of water drop, dynamic change in contact angle, water vapor permeability (WVP), and water absorptivity were also affected by biopolymer coating, but the degree of change was dependent on the coating materials as well as treatment methods. For example, the rate of change in contact angle of water on paperboards decreased dramatically from 1.4 to 3.8 times depending on the coating materials as well as treatment methods. Generally, SPI-coated paperboards were more water-resistant than alginate coated ones. However, water resistance of the alginate-coated paperboards post-treated with the $CaCl_2$ solution was comparable to the SPI coated ones.
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1. Introduction

Corrugated fiberboard containers are the most widely used packaging materials for the shipping and handling of food and nonfood materials. It has the highest volume of any single packaging material in the world and represents more than 80% of the volume of all paper-based packaging materials (Twede & Selke, 2005). Corrugated fiberboard, which is composed of the linerboard and the medium, is characterized by its cellular structure, which imparts high compressive strength at a relatively low weight. Though corrugated board is the most highly recyclable packaging material, due to its composition of cellulose, the hydrophilic nature of cellulose causes it to be less water resistant.

Paperboards are prone to absorb water vapor from the environment, especially when stored under high humidity conditions or when coming into contact with high-moisture food materials such as fresh agricultural produce. Absorption of moisture reduces physical and mechanical strength of the paperboards, causing corruption of boxes during storage and distribution. Surface treatments such as sizing and coating are usually used to improve physical strength as well as water barrier properties.

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A water barrier can be formed by changing the wettability of the paper surface with sizing agents or through coating with hydrophobic materials. Paraffin wax applied in a molten form was commonly used to produce a water vapor barrier, but polyethylene (PE) applied by extrusion gives a more durable and flexible coating on paper or paperboard (Robertson, 1993). However, the content of the PE and paraffin of packaging materials makes it difficult to separate, recycle, or compost them after use (Robson, 1993).

In an effort to produce more environmentally friendly materials, renewable and biodegradable biopolymers have been investigated as paper coating materials (Han & Krochta, 1999, 2001; Lin & Krochta, 2003; Park et al., 2000; Rhim, Hwang, Park, Kang, & Jung, 1998; Trezza & Vergano, 1994). Trezza and Vergano (1994) reported zein-coated papers were as effective as PE laminated paper used for quick-service restaurant sandwich packaging. Rhim et al. (1998) showed that the grease resistance of carrageenan-coated papers was comparable to PE laminated papers, and Park et al. (2000) reported that soy protein-coated papers imparted gas and lipid barrier, as well as adequate mechanical properties, for extending shelf life of food products. Whey proteins (Han & Krochta, 2001; Lin & Krochta, 2003) were also tested for the preparation of protein-coated papers to improve lipid barrier property as well as physical and surface color properties. Han and Krochta (1999) showed that whey protein isolate (WPI) coating on pulp paper increased ink printability and reduced water vapor permeability (WVP). Most of the research works on biopolymer-coated papers has been focused on improving lipid barrier properties of paper mainly because most of biopolymer films are hydrophilic, they would be expected to be good lipid barriers at low relative humidity. However, attempts on improving water barrier properties of paper packaging materials with biopolymer coating have been scarcely reported. Our previous research showed that alginate-based films cross-linked with Ca ions by post-treatment of $CaCl_2$ solution (Rhim, 2004; Rhim, Kim, & Kim, 2003), soy protein films cross-linked by formaldehyde post-treatment (Rhim, 1998; Rhim & Weller, 2000), and soy protein films composited with layered silicates such as montmorillonite (Rhim, Lee, & Kwak, 2005) increased water barrier properties of such biopolymer-based films.

The main objective of this study was to test the effect of biopolymer coating, such as alginate and soy protein, on the physical strength and water resistance properties of paperboard used as liner board in corrugated fiberboard boxes. The effect of property modification of the biopolymers such as ionic cross-linking of alginate film by $CaCl_2$ treatment, cross-linking of soy protein film by formaldehyde treatment, and compositing alginate or soy protein with a layered silicate (montmorillonite) was also investigated.

2. Materials and methods

2.1. Materials

SC manila paperboard (SK180, basis weight: 180 g/m^2) was obtained from Kyungnam Paper Manufacturing, Co., Ltd. (Yongin, Korea). Soy protein isolate (SPI) (minimum 90 g/100 g protein content on dry basis, SUPRO 620) was obtained from Protein Technologies International (St. Louis, MO, USA) and sodium alginate was purchased from Kanto Chemical Co., Ltd. (Tokyo, Japan). Organically modified montmorillonite (OMMT; Namomer I.34TCN) was purchased from Nanocor Inc. (Arlington Heights, IL, USA).

2.2. Preparation of coating solution

The alginate coating solution was prepared with the same method used for alginate film preparation (Rhim, 2004; Rhim et al., 2003). Five grams of alginate powder was added to a constantly stirred mixture of 100 ml of distilled water with 2 g of glycerol as a plasticizer. The mixture was heated on a hot plate with constant stirring until completely dissolved and clear. The SPI coating solution was prepared based on the method of SPI film preparation (Rhim, 1998; Rhim & Weller, 2000). After dissolving 15 g of SPI into 100 ml of distilled water, 6 g of glycerol was added as a plasticizer. SPI film solutions were prepared without pH adjustment.

Composite coating solutions of alginate or SPI with OMMT were prepared in accordance with the method of composite film preparation (Rhim, Lee, & Kwak, 2005). In each solution, OMMT at a ratio of 0.2 g OMMT/g solid was dissolved in due amount of distilled water and glycerol by stirring for 30 min. A predetermined amount of alginate or SPI was then added and dissolved while stirring on hot plates. After dissolving, all the film preparation solutions were heated for 20 min at 90 °C in a constant-temperature water-bath.

2.3. Paper coating

Coating solutions prepared as above were coated onto SC manila paperboard (20 × 27 cm) using wire bar coaters (SA-203 Wire Bar Coaters, Donghak Machine Co., Ltd., Korea). Paperboard was mounted on a flat glass plate by applying masking tape around the paper edges to prevent wrinkling of paper during coating. A spoonful of each coating solution was loaded on the prepared paper and spread over the paper using a No. 20 wire bar coater. A wire bar coater, originally called a Mayer Rod, and also referred to as a wire doctor equalizer, or metering rod, is a round steel bar that is wrapped very tightly with wire. Generally, film thickness is determined by the thickness of a wire used, which is classified as a number. The higher the number,

the thicker the film coating is made. The coated papers were dried under ambient conditions for 24 h.

Half of each alginate- or SPI-coated paperboards were modified with $CaCl_2$ or formaldehyde treatments. For the modification of alginate-coated paper, the surface of the alginate-coated side was immersed with a solution (5 g $CaCl_2$/100 ml distilled water) for 3 min. The solution was then removed and the paper dried at room temperature. SPI-coated paper was treated with formaldehyde vapor for 2 h in a desiccator which was saturated with formaldehyde.

2.4. Conditioning

All the paperboards were cut for property testing and conditioned in a constant-temperature humidity chamber (model FX 1077, Jeiotech Co., Ltd., Korea) set at 25 °C with 50% RH for at least 48 h to adjust the moisture content.

2.5. Moisture content and thickness measurements

Moisture content of test samples was measured using the oven drying method (105 °C for 24 h). Thickness of paper samples was measured using a hand-held micrometer (Dial thickness gauge No. 7301, Mitutoyo Co. Ltd., Japan) with a 0.01-mm accuracy. Five thickness measurements were taken on each sample along the length of the strip, mean values were reported.

2.6. Tensile strength (TS) and percentage elongation at break (E)

TS and E of each paper sample were evaluated with a Model 4465 Instron Universal Testing Machine (Instron Engineering Corporation, Canton, MA, USA) in accordance with the ASTM method D882-97 (1999). Tensile testing samples were 2.54 cm × 10 cm rectangular strips. Initial grip separation was set at 50 mm and cross-head speed was set at 500 mm/min. TS was calculated by dividing the maximum load by the initial cross-sectional area of the paper. E was calculated by dividing the extension at rupture of the paper by the initial length of the sample (50 mm) and multiplied by 100. TS and E measurements for each type of paper were replicated three times with individually prepared papers as the replicated experimental units, each replicate being the means of seven tested sampling units taken from the same paper.

2.7. Ring crush test

All paper samples were cut into strips (12 mm × 153 mm) using a ring crush cutter. Samples were then divided into two groups: one stored at room temperature and the other stored at 30 °C and 98% RH for 5 days prior to the tests. Each strip of samples was placed in the crush tester and the ring crush strength was measured using a computer-controlled Advanced Universal Testing System (model LRX*Plus*, Lloyd Instrument Limited, Fareham, Hampshire, UK) at room temperature in accordance with TAPPI Test Method T 822 om-93. A 500-Newton (N) load cell was used, and the cross-head speed was 50 mm/min. At least five specimens for each paper sample were tested, and their mean values were reported.

2.8. Water vapor permeability

WVP was measured by modified ASTM method E96-00 (2002) using a WVP cup (46 mm of ID and 27 mm in depth) made of poly(methyl methacrylate). Paper samples cut into 7 cm × 7 cm were mounted on the WVP cups to expose the coated face to the outside of the WVP cups filled with 18 ml of distilled water. The cups were placed in an environmental chamber (Model FX 1077, Jeio Tech Co. Ltd., Ansan, Korea) set at 25 °C and 50% RH. A fan was operated within the chamber creating an air velocity of 198 m/min over the surface of the cups to circulate the air in the chamber. Weight of cups was monitored by weighing every hour for a period of 8 h. Slopes of the steady-state portion of weight loss versus times curves were determined by linear regression to estimate water vapor transmission rate (WVTR). The WVP (ng m/m^2 s Pa) was calculated as

$$\mathrm{WVP} = (\mathrm{WVTR}\, L)/\Delta p,$$

where WVTR was the measured water vapor transmission rate (ng/m^2 s) through a paper, L was the mean paper thickness (m) and Δp was the partial water vapor pressure difference (Pa) across the two sides of the paper. In calculating WVP, the effect of resistance of the stagnant air layer between the paper's underside and the surface of the water in the cups was corrected for by the method of Gennadios, Weller, and Gooding (1994).

2.9. Contact angle

Contact angle of water in air on the paper surface was measured with a contact angle analyzer (model Phoenix 150, Surface Electro Optics Co. Ltd., Kunpo, Korea) after a water drop of ca. 10 μl was placed on the surface of coated or uncoated paper using a micro-syringe. All paper samples (3 cm × 10 cm) were glued on the movable sample stage (black Teflon-coated steel, 7 cm × 11 cm) leveled horizontally before measurement. The contact angle on both sides of the drop was measured to assume symmetry and horizontal level. Distilled water was used as a probe liquid. Dynamic contact angle change was measured by recording the contact angle change of a water drop with time within 400 s at room temperature at 50 ± 5% RH.

2.10. Water absorptivity

Water absorptivity of the paper samples was determined using the method and apparatus described by Han and Krochta (1999). A paper sample (7 cm × 7 cm) was placed with the coating face up between the base and the ring of the apparatus and firmly clamped with a screw nut. Five milliliters of water was applied to the surface of the paper and placed on a flat lab bench under room conditions (20±2 °C) to allow water absorption with time. After absorption of water for a predetermined time, water was discarded and the excess surface water on the paper was removed by wiping with tissue paper. To get rid of the edge effect, the paper was cut into a 4 cm × 4 cm rectangles with a paper cutter. The absorptivity (A; g water/m^2) was determined as

$$A = (W_2 - W_1)/S$$

where W_1 and W_2 are the weights of the paper before and after water absorption, respectively, and S is the surface area of the paper sample (0.0016 m^2).

2.11. Statistical analysis

All the test property measurements of each sample were triplicated with individually prepared coated or noncoated paperboard as the replicated experimental units. Statistics on a completely randomized design were determined using the General Linear Models procedure using the Statistical Analysis System (SAS Inst. Inc., Cary, NC, USA). Mean property values were separated ($P<0.05$) with the Duncan's multiple range test.

3. Results and discussion

3.1. Coated paperboards

All the alginate- or SPI-coated paperboards had a smooth surface after coating. Wrinkling of paperboards during coating and drying was prevented by fixing the paper on a glass plate by taping around the edges. This indicates that the concentration of coating solutions (i.e., 5 g alginate/100 ml distilled water and 15 g SPI/100 ml distilled water) was suitable for coating on the paperboard used. Han and Krochta (1999) showed that 5 g/100 ml WPI solution could not generate a homogeneous coating on paper due to the excess amount of water, which resulted in a wrinkled paper. By increasing the concentration up to 10 g/100 ml, they could get a homogeneously coated paper. Coating increased the thickness of the paperboards from 9% to 16% depending on the coating materials (Table 1). Generally, coat thickness is influenced by the solid content of the coating solution (Han & Krochta, 1999), though it is not proportional to the solid content. Both alginate and SPI coating solutions blended with OMMT resulted in a thicker coat due to the higher solid content.

It is interesting to note that biopolymer coating induced a more homogeneous surface on paperboards, which is evidenced by the lower standard deviations in the thickness of coated paperboards. Biopolymer coating decreased the standard deviations of the thickness measurement by two or three fold. This effect is mainly due to the fact that the pores of the cellulose structure of the paper are filled with coating materials. Han and Krochta (2001) demonstrated through SEM image analysis that WPI coating produced a more homogeneous and smooth coating surface than its uncoated counterpart. In this case, the coating materials acted in a

Table 1
Tensile strength (TS) and elongation at break (E) of biopolymer-coated paperboards[1]

Paperboards	Paper thickness (μm)	Thickness increase (μm)	TS (MPa)	E (%)
Uncoated[2]	236.7±6.0^{a}	0	40.0±3.1^{d}	3.9±0.4^{a}
Alg[3]	257.8±3.2^{b}	21.1	33.6±1.4^{bc}	4.1±0.2^{ab}
Alg/$CaCl_2$[4]	264.2±3.1^{cd}	30.3	35.3±2.6^{cd}	4.6±0.2^{ab}
Alg/OMMT[5]	267.0±3.4^{d}	25.0	34.2±2.9^{bcd}	4.7±0.2^{ab}
SPI[6]	261.7±2.1^{bc}	27.5	25.0±7.9^{a}	6.0±2.2^{ab}
SPI/FA[7]	261.0±3.2^{bc}	24.3	26.0±5.6^{ab}	6.9±1.9^{b}
SPI/OMMT[8]	275.3±1.9^{e}	38.6	27.0±5.9^{abc}	6.6±2.6^{ab}

[1]Means of three replicates±standard deviations. Any two means in the same column followed by the same letter are not significantly ($P>0.05$) different by Duncan's multiple range tests.
[2]Uncoated paperboards.
[3]Alginate film-coated paperboards.
[4]Alginate film-coated and $CaCl_2$-treated paperboards.
[5]Alginate and OMMT composite film-coated paperboards.
[6]SPI film-coated paperboards.
[7]SPI film-coated and formaldehyde-treated paperboards.
[8]SPI and OMMT composite film-coated paperboards.

similar way to fillers, like clay. Fillers fill in the void areas on the surface of the paper to provide a smooth surface. The smoothness provides for good printing resolution and reduces ink consumption.

3.2. Tensile strength (TS) and elongation at break (E)

Table 1 also shows the results of tensile testing. TS is one of the most basic strength properties of packaging materials. TS is a measure of the ability of paperboard to resist breaking under tension, which is dependent on the strength of fibers, their surface area and length, and the bonding strength between them. TS decreased significantly ($P<0.05$) as a result of biopolymer coating, while E increased slightly. The change in tensile properties from coating was dependent on the coating materials. The changes in tensile properties were more significant in the SPI-coated group than the alginate-coated group. This result indicates that the biopolymer coating increased the flexibility of the paperboard. Han and Krochta (2001) also found that the tensile modulus of paper was decreased, and the breaking strain (in % elongation) increased with WPI coatings. They also found that the breaking force decreased with increasing coating weight, while the breaking strain was not changed. The decrease in TS of coated paperboards is mainly due to swelling of cellulose fiber by solvent penetration during coating and may be partially due to the fact that coating materials are impregnated into the cellulose structure of paper and interfere with fiber-to-fiber interaction. Han and Krochta (2001) also suggested that such interference by the coating materials causes a reduction of interaction force between the fibers of coated papers, consequently resulting in decreased TS of coated paper.

3.3. Ring crush strength

Though TS is an important factor in package performance where boxes need to support weight, compression strength is a more crucial factor with corrugated container boxes. Usually a ring crush test is used as the standard test for the linerboards of corrugated fiberboard boxes. Table 2 shows the ring crush test results of paperboards. Basically, the ring crush strength of the paperboard was not affected by biopolymer coating. There was no significant ($P<0.05$) difference in ring crush force between control groups and biopolymer-coated groups right after coating. However, ring crush force of all the paperboards decreased appreciably after 5 days storage at 30 °C and 98% RH condition. This decrease is mainly caused by the increase in the moisture content of paperboards. It is notable that the degree of reduction varies depending on the coating materials. This result shows that the degree of water absorption changed depending on the coating material and had an influential effect on the ring crush force of the paperboards.

3.4. Water vapor permeability (WVP)

The WVP of uncoated paperboards was 3.6 ± 0.02 ng m/m^2 s Pa (Table 3). Biopolymer coating affected WVP of paperboards in a different way depending on the polymer materials, and post-treatment method, and OMMT blending. The coating material was most influential to the WVP of biopolymer-coated paperboards. Alginate coating increased WVP slightly, while SPI coating reduced WVP significantly ($P<0.05$). This seems to be contradictory to the previous findings on WVP values of Alginate and SPI films. The WVP of

Table 3
Moisture content (MC) and water vapor permeability (WVP) of biopolymer-coated paperboards

Paperboards	MC (g H_2O/100 g)	WVP (ng m/m^2 s Pa)
Uncoated	4.5 ± 0.4^{c}	3.6 ± 0.2^{c}
Alg	3.5 ± 0.6^{ab}	3.9 ± 0.1^{d}
Alg/$CaCl_2$	4.1 ± 0.4^{bc}	3.7 ± 0.1^{cd}
Alg/OMMT	3.3 ± 0.4^{a}	3.6 ± 0.1^{c}
SPI	3.6 ± 0.2^{bc}	3.1 ± 0.1^{b}
SPI/FA	3.5 ± 0.1^{ab}	2.3 ± 0.1^{a}
SPI/OMMT	3.7 ± 0.3^{ab}	2.9 ± 0.2^{b}

Table 2
Change in moisture content and ring crush force of biopolymer-coated paperboards during storage

Paperboards	MC (g H_2O/100 g solid)		Ring crush force (N)	
	Day_1	Day_5	Day_1	Day_5
Uncoated	3.6 ± 0.2^{a}	8.6 ± 0.2^{a}	188.0 ± 6.2^{a}	140.9 ± 3.9^{d}
Alg	$4.6\pm0.3b^{cd}$	10.4 ± 0.5^{c}	201.8 ± 25.9^{a}	137.1 ± 10.7^{d}
Alg/$CaCl_2$	5.2 ± 0.1^{e}	12.1 ± 0.4^{e}	180.7 ± 14.4^{a}	98.8 ± 3.6^{a}
Alg/OMMT	5.1 ± 0.6^{de}	9.5 ± 0.3^{b}	195.8 ± 8.2^{a}	139.6 ± 6.4^{d}
SPI	4.0 ± 0.6^{ab}	10.7 ± 0.6^{cd}	186.9 ± 19.5^{a}	115.0 ± 5.9^{b}
SPI/FA	4.2 ± 0.1^{abc}	10.6 ± 0.1^{cd}	194.3 ± 6.8^{a}	118.2 ± 5.9^{bc}
SPI/OMMT	4.7 ± 0.1^{cde}	11.1 ± 0.1^{d}	182.3 ± 7.3^{a}	130.0 ± 8.7^{cd}

alginate (Rhim, 2004) and SPI (Rhim, 1998) films was reported to be 1.42 ± 0.02 and 1.96 ± 0.03 ng m/m^2 s Pa, respectively. Han and Krochta (1999) reported that WVP of 100% pulp paper (thickness: 188 ± 2.1 μm) and 10% WPI-coated paper (thickness: 196 ± 1.2 μm) were 2.5 ± 0.05 and 1.4 ± 0.05 ng m/m^2 s Pa, respectively. Though their values are slightly lower than the results in the present study, they are practically in the same range. The different effect of each biopolymer on WVP of coated papers seems to be due to the difference in interaction of biopolymers with the structural cellulose fiber of the paper. In addition, the effect of post-treatment or blending with OMMT is different in both biopolymers. SPI coatings with formaldehyde post-treatment or OMMT compositing were more effective in decreasing the WVP of coated paperboards.

3.5. Initial contact angle of water

Results of the initial contact angle measurement of paperboards with or without biopolymer coatings were summarized in Table 4. The initial contact angle of water on uncoated paperboard was $56.3 \pm 3.8^\circ$, which is much lower than the result reported by Han and Krochta (1999). Their reported value was about 100°. This discrepancy can be mainly attributed to the fact that they used different paper materials with a different composition, i.e., a virgin pulp paper. In addition, these materials had different surface characteristics such as surface smoothness and homogeneity. Coating with biopolymer materials reduced the contact angle of water significantly ($P<0.05$), up to the level of 35.3–48.6° depending on the biopolymer and the post-treatment method. Reduction in the contact angle of water by biopolymer coatings indicates an increase in hydrophilicity of the surface of the paperboards. A more hydrophilic surface of paperboards results in a higher wetting energy, a lower spreading coefficient, and a higher work of adhesion as shown in Table 4. Decrease in contact angle of water on biopolymer-coated paperboards can be also explained by the fact that the surface of the paperboards became smoother and more homogeneous by filling the rough surfaces or pores of the paperboards with hydrophilic biopolymers resulting in an increased affinity of the paperboards to water. Han and Krochta (1999) also found a decrease in contact angle of water by coating paper with WPI. As a whole, the contact angle of water on the alginate-coated paperboards decreased more than that of the SPI coated ones. This indicates that the water resistance of SPI-coated paperboards is higher than that of alginate-coated paperboards. This result supports the findings of the WVP measurement which showed a lower WVP of the SPI-coated paperboards (Table 3). In addition, it is worthwhile to note that the degree of change in contact angle of water of each treatment is consistent with that of the WVP measurement.

3.6. Dynamic change of contact angle

Fig. 1 shows the result that dynamic change of contact angle of water on the paperboards. The contact angle of

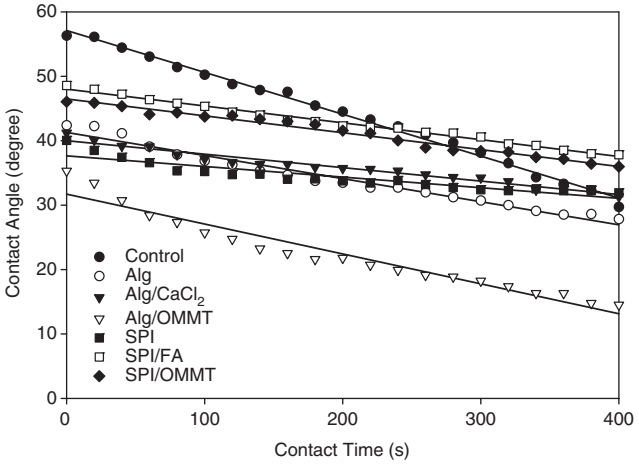


Fig. 1. Dynamic change in contact angle of water drop on biopolymer-coated paperboards. (●) Uncoated, (○) Alg, (▼) Alg/$CaCl_2$, (△) Alg/OMMT, (■) SPI, (□) SPI/FA, (◆) SPI/OMMT.

Table 4
Results of contact angle measurement of biopolymer-coated paperboards

Paperboards	Avg contact angle (deg)	Left angle (deg)	Right angle (deg)	Wetting energy[1] (mJ/m^2)	Spreading coeff.[2] (mN/m)	Work of adhesion[3] (mJ/m^2)
Uncoated	56.3 ± 3.8^{d}	56.1 ± 5.8^{e}	56.5 ± 6.6^{e}	40.3 ± 4.0^{a}	32.5 ± 4.0^{e}	113.1 ± 4.0^{a}
Alg	42.4 ± 3.2^{b}	42.7 ± 5.9^{bc}	42.1 ± 6.5^{bc}	53.7 ± 2.8^{cd}	19.1 ± 2.8^{bc}	126.5 ± 2.8^{cd}
Alg/$CaCl_2$	41.2 ± 5.7^{b}	43.1 ± 6.6^{bc}	39.3 ± 7.5^{b}	54.5 ± 4.8^{d}	18.3 ± 4.8^{b}	127.3 ± 4.8^{d}
Alg/OMMT	35.3 ± 3.2^{a}	36.2 ± 3.4^{a}	34.4 ± 5.0^{a}	59.3 ± 2.4^{e}	13.5 ± 2.4^{a}	132.1 ± 2.4^{e}
SPI	40.1 ± 2.8^{b}	39.2 ± 3.2^{ab}	41.0 ± 3.5^{bc}	55.7 ± 2.3^{d}	17.1 ± 2.3^{b}	128.5 ± 2.3^{d}
SPI/FA	48.6 ± 2.9^{c}	48.5 ± 4.0^{d}	48.6 ± 2.2^{d}	48.1 ± 2.7^{b}	24.7 ± 2.8^{d}	120.9 ± 2.8^{b}
SPI/OMMT	46.1 ± 1.1^{c}	47.0 ± 1.6^{cd}	45.1 ± 2.7^{cd}	51.1 ± 1.6^{c}	21.7 ± 1.6^{c}	123.9 ± 1.6^{c}

[1]Wetting energy = surface tension of a probe liquid (72.8 mJ/m^2 for water) × cos θ, where θ is an average contact angle measured.
[2]Spreading coefficient = surface tension of a probe liquid × (cos θ − 1).
[3]Work of adhesion = surface tension of a probe liquid × (cos θ + 1).

a water drop on biopolymer-coated or uncoated paperboards decreased linearly over a short time period. This result clearly indicates the difference in initial contact angle of water on each paperboard and the difference in the degree of decrease of contact angle with time. The contact angle of water on the paperboards decreases due to water absorption by the paperboard or evaporation of water into the atmosphere. Generally, in a short time span, evaporation of water is slower than water absorption by the paperboard. Consequently, the change in contact angle with time can be used as an indication of the water absorption rate by the paperboards (Han & Krochta, 1999). The linear regression result revealed that the linear model matched the dynamic contact angle change data with high values of the coefficient of determination (R^2) as shown in Table 5.

Table 5 also shows the rate of contact angle change of water on each paperboard. The rate of contact angle change on uncoated paperboard was $0.065 \pm 0.0011^{\circ}$/s. It decreased dramatically with biopolymer coating, a decrease of 1.4–3.8 times depending on the coating material and treatment method. A lower decrease in the contact angle change rate (k) indicates greater water resistance of the paperboards. Generally, a greater decrease in k was found in the SPI-coated paperboards than in the alginate coated ones, with the exception of the alginate-coated and $CaCl_2$-treated paperboards. Again, this trend supports the previously measured WVP and initial contact angle results.

3.7. Water absorptivity

The water resistance of the biopolymer-coated paperboards was measured through direct contact with the surface of the boards with water (Fig. 2). The water absorptivity of the paperboards increased as water contact time increased. However, the degree of increase in water absorptivity was quite different depending on the coating materials and post-treatment method. The water absorptivity of paperboards coated with alginate, or alginate with OMMT increased considerably compared with the other groups, whose water absorptivity was comparable to or even lower than uncoated one. Higher water absorptivity of paperboards coated with alginate or alginate with OMMT was mainly caused by the fact that the coating materials themselves rather than paperboards absorbed water and swelled considerably. It is interesting to note that this trend supports the other wetting property measurements of the coated paperboards. Han and Krochta (1999) reported that the water absorptivity of paper increased by coating with WPI and they also reported that the thicker the coating, the higher the increase in water absorptivity. They concluded that WPI coating accelerated the water absorption and elevated the total water absorption capacity and therefore the WPI may work as a binding agent as well as a surface sizing agent after coated on the paper. However, the result of the present research indicates that by choosing appropriate biopolymer materials or proper modification methods, the water resistance of paperboards can be improved.

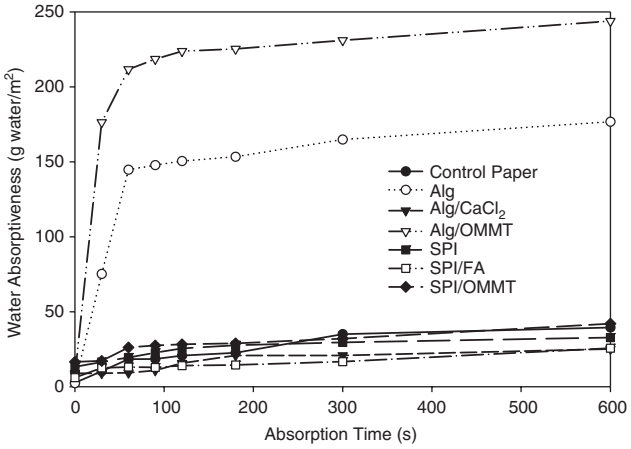


Fig. 2. Change in water absorptivity of biopolymer-coated paperboards. (●) Uncoated, (○) Alg, (▼) Alg/$CaCl_2$, (△) Alg/OMMT, (■) SPI, (□) SPI/FA, (◆) SPI/OMMT.

Table 5
Contact angle decreasing rate (k) and initial contact angle of biopolyme- coated paperboards estimated from linear regression analysis

Paperboards	k (deg/s)	Initial contact angle (deg)	R^2
Uncoated	0.065±0.001	57.1±0.3	0.99
Alg	0.036±0.002	41.3±0.3	0.97
Alg/$CaCl_2$	0.021±0.001	40.0±0.2	0.96
Alg/OMMT	0.046±0.003	31.7±0.6	0.94
SPI	0.017±0.002	37.6±0.4	0.85
SPI/FA	0.026±0.001	48.0±0.2	0.99
SPI/OMMT	0.026±0.001	46.5±0.2	0.98

4. Conclusion

Mechanical and water barrier properties of paperboards can be modified by biopolymer coatings. Though TS of paperboards was decreased by biopolymer coating, ring crush force was not decreased. Water barrier properties were increased by biopolymer coating especially for SPI or alginate coating with $CaCl_2$ post-treatment. This result indicates that water barrier properties of paperboards can be improved by the proper choice of biopolymer material and coating method. The improved water barrier properties of the paperboards can be exploited in the preparation of water-resistant corrugated fiberboard boxes for the storage and distribution of high-moisture foods such as fresh agricultural produce.

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